AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL **Submission:** model_2

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (`docs/markschemes.pdf`).

1. Final mark

Final mark (choose one): $\boxed{7}$

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. `ME1`) where possible.

A complete and correct solution. It proves the common residue modulo g, establishes $\gcd(r, g) = 1$, obtains the bound $r + g < 10^6$ for $k = 3$ and $g \leq 500000$ for $k \geq 4$, reduces to the condition $r^2 \equiv -1 \pmod{g}$, rules out $g > 998285$, and exhibits a valid construction attaining $g = 998285$. The only blemish is a trivial omission (not stating $k \geq 2$ when excluding the residue $r = 0$), which does not affect the mark. Scores **7**.